

Predicting New-User Engagement on Zindi

October 7, 2026

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Will a user who just signed up still be active in their second month?

Data: [Zindi New User Engagement Prediction Challenge](#) (Zindi, 2022–23)

Outline

1. Problem & data (load, build target, X / y)
2. Data preparation: encoding, missing values, feature engineering
3. Exploratory analysis with Seaborn $\rightarrow$ what it changes in the ML setup
4. Validation design & metric
5. Models, hyper-parameter search, results
6. Findings, real-world impact, conclusions

1. The problem

Zindi is Africa’s largest data-science competition platform (≈ 12 k new sign-ups in the 7 months of this dataset). Most sign-ups are *one-month tourists*: in our data only **12–21** % of a typical monthly cohort comes back in its second month.

Task. For every user who registered in month m , predict whether they will do **anything** on Zindi (view a page, join a competition, submit, post...) in month $m + 1$. $\rightarrow$ binary classification.

Why it matters

- Acquisition is expensive; a user who survives month 2 is far more likely to become a long-term community member, competitor and potential hire for Zindi’s clients.
- Churn happens fast. Zindi needs to know **at the end of the first month** who is about to drop out, while there is still time to act.

How the model helps

- *Targeted retention*: send onboarding nudges / beginner competitions / mentorship to users with low predicted probability instead of spamming everybody $\rightarrow$ lower cost, less unsubscribe fatigue.
- *Prioritisation*: identify likely-engaged newcomers for community programmes and hackathon invitations.
- *Product insight*: which first-month behaviours drive retention (e.g. submitting vs. only browsing)?

1.1 Loading the data with pandas

Eight relational tables, linked by `User_ID` / `Competition ID`. Dates are **masked**: year and month are recoded (“*months are in chronological order but January is not necessarily month 1*”), time of day and day-of-month are kept.

```
[2]: D = '../data/' # raw Zindi tables (shared by both
      ↪versions)
users = pd.read_csv(D + 'Users.csv')
act = pd.read_csv(D + 'UserActivity.csv') # page views / clicks
part = pd.read_csv(D + 'CompetitionParticipation.csv') # competition enrolments
disc = pd.read_csv(D + 'Discussion.csv')
com = pd.read_csv(D + 'Comments.csv')
comps = pd.read_csv(D + 'Competition.csv')
sample_sub = pd.read_csv(D + 'SampleSubmission.csv')

pd.DataFrame({n: [len(t), t.shape[1]] for n, t in
              dict(Users=users, UserActivity=act, CompetitionParticipation=part,
                  Discussion=disc, Comments=com, Competition=comps).items()},
              index=['rows', 'cols']).T
```

```
[2]:
```

	rows	cols
Users	12413	8
UserActivity	317292	6
CompetitionParticipation	8385	8
Discussion	1439	9
Comments	467	6
Competition	247	20

```
[3]: users.head(3).T # transposed: one column per user
```

```
[3]:
```

	0	1	2
FeatureX	0	1	0
FeatureY	0	0	0
User_ID	ID_DC6S4E90	ID_E8S97OUT	ID_QZ1HASL3
Countries_ID	X9GR	X9GR	X9GR
Created At time	19:33:13.663391	22:18:18.228921	23:13:03.266635
Created At Year	1	1	1
Created At Month	1	12	1
Created At Day_of_month	16	27	1

```
[4]: act.head(3).T
```

```
[4]:
```

	0	1	2
User_ID	ID_RT43AK77	ID_RT43AK77	ID_RT43AK77
Title	comp_ID_IV5D	comp_ID_IV5D	comp_ID_IV5D
datetime time	22:00:38	00:41:02	00:53:15
datetime Year	1	1	1
datetime Month	11	11	11
datetime Day_of_month	22	23	23

1.2 Recovering the time axis

All users are in masked *Year 1*, with masked months {11, 12, 1, 2, 3, 4, 5}. In what order? We cross-tabulate **sign-up month** (rows) vs **activity month** (columns): users can only be active *after* they register, so the zero pattern reveals the chronology.

```
[5]: ORDER = {11: 0, 12: 1, 1: 2, 2: 3, 3: 4, 4: 5, 5: 6} # masked month ->
      ↪chronological index t
LABEL = {v: f'M{k}' for k, v in ORDER.items()} # t -> display label
```

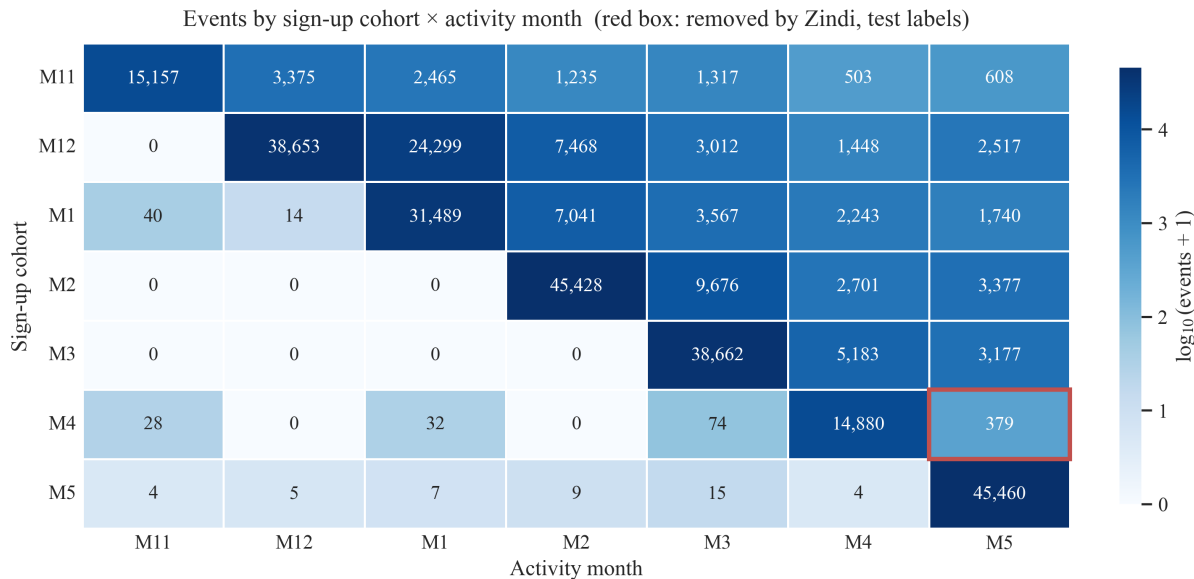
```

def standardise(df, prefix):
    df = df.rename(columns={f'{prefix} Month': 'month', f'{prefix} Day_of_month': 'day',
    ↪f'{prefix} time': 'time'})
    df['t'] = df['month'].map(ORDER)
    return df

users = standardise(users, 'Created At').rename(columns={'t': 'cohort', 'day': 'signup_day'})
act, part = standardise(act, 'datetime'), standardise(part, 'Created At')
disc, com = standardise(disc, 'Created At'), standardise(com, 'Created At')

am = act.merge(users[['User_ID', 'cohort']], on='User_ID')
ct = pd.crosstab(am.cohort, am.t.rename('activity month'))
ct = ct.rename(index=LABEL, columns=LABEL)
fig, ax = plt.subplots(figsize=(9.5, 4.6))
sns.heatmap(np.log10(ct + 1), annot=ct.apply(lambda col: col.map('{:,}'.format)), fmt='',
    ↪cmap='Blues',
    linewidths=0.8, linecolor='white', annot_kws={'size': 11},
    cbar_kws={'label': r'$\log_{10}$(events + 1)', 'shrink': 0.9}, ax=ax)
r_test, c_test = list(ct.index).index('M4'), list(ct.columns).index('M5') # month-2
    ↪activity of the test cohort
ax.add_patch(plt.Rectangle((c_test, r_test), 1, 1, fill=False, edgecolor=RED, lw=2.5,
    ↪clip_on=False))
ax.set(xlabel='Activity month', ylabel='Sign-up cohort',
    title='Events by sign-up cohort × activity month (red box: removed by Zindi, test
    ↪labels)')
ax.tick_params(axis='both', length=0); ax.tick_params(axis='y', rotation=0); ax.grid(False)
plt.show()

```



Reading the matrix

- (Almost) upper-triangular — only a few dozen stray events below the diagonal $\Rightarrow$ chronological order is **M11 $\rightarrow$ M12 $\rightarrow$ M1 $\rightarrow$ M2 $\rightarrow$ M3 $\rightarrow$ M4 $\rightarrow$ M5** (7 consecutive months).
- The **M4 cohort** has essentially no M5 activity (8 users out of 1 382): Zindi removed it — the M4 cohort is the hidden **test set** (`SampleSubmission` = 1 340 M4 users, “_Month_5”).

- $\Rightarrow$ We must **build the training labels ourselves** from cohorts M11 ... M3 (their month- $m+1$ is observed).

1.3 Target definition and X / y split

Active = 1 if the user has **any** record (activity log, competition enrolment, discussion or comment) in the month right after the sign-up month. Features may only use information from the **sign-up month** (what Zindi knows at the moment it has to act) — anything later would leak the target.

```
[6]: events = pd.concat([x[['User_ID', 't']] for x in (act, part, disc, com)])
ev = events.merge(users[['User_ID', 'cohort']], on='User_ID')
active_next = set(ev.loc[ev.t == ev.cohort + 1, 'User_ID'])
users['Active'] = users.User_ID.isin(active_next).astype(int)

LABELLED = [0, 1, 2, 3, 4]      # M11..M3 cohorts: label observed
TEST_COHORT = 5                # M4 cohort: label hidden by Zindi
summary = users[users.cohort <= TEST_COHORT].groupby('cohort').agg(users=('Active', 'size'),
    ↪active_rate=('Active', 'mean'))
summary.index = summary.index.map(LABEL); summary.loc['M4', 'active_rate'] = np.nan
summary.T.round(3)
```

```
[6]: cohort      M11      M12      M1      M2      M3      M4
users      1262.000  1982.000  1504.000  2278.000  2000.000  1382.0
active_rate    0.198    0.784    0.207    0.175    0.122    NaN
```

2. Data preparation

Issue	What we do	Why
Events are in long, multi-table logs	aggregate per user within the sign-up month only	one row per user; no look-ahead leakage
70+ raw event titles (comp_ID_xxx, badge_xxx, ...)	group into 18 behaviour families (view competition, submit, download data, ...)	IDs are too sparse; the <i>type</i> of behaviour is what generalises
Countries_ID missing for 47 % of users	category Missing + binary flag	missingness is informative (see EDA) — imputing a country would be invented data
146 masked countries (high cardinality)	keep top-10, rest $\rightarrow$ Other , one-hot	enough users per level; avoids over-fitting rare countries
FeatureY $\in \{0, 1, 3\}$ (masked category)	one-hot	codes are nominal, not ordinal
Successful Submission Count = strings "count 10", NaN	parse number; NaN $\rightarrow$ 0	NaN = no successful submission
Users with no events in a table	counts $\rightarrow$ 0	absence of activity is a true zero
Late sign-ups have less time to show activity	days_observed, active_day_ratio, days_since_last	normalises activity by exposure

Issue	What we do	Why
Month lengths are masked too (M2 has 31 days!), M4 log stops at day 22	window end = last logged day of that month	keeps exposure features consistent between train and test

Engineered features (all computed from the sign-up month)

- **Volume / regularity:** `n_events`, `n_active_days`, `n_distinct_actions`, `n_distinct_comps`, `active_day_ratio`
- **Recency:** `days_since_last` = days between last activity and month end (fresh activity $\Rightarrow$ likely to continue)
- **Behaviour mix:** counts for 18 event families (submissions, data downloads, learning pages, jobs ...)
- **Competition context** (from `CompetitionParticipation` + `Competition`):
 - `n_comps_joined`, `max_sub_bucket` (masked successful-submission bucket)
 - `n_new_comps_joined`: competitions whose **first appearance** in the logs is in the sign-up month — a freshly launched competition is likely still running next month (computed from past/present logs only)
 - `joined_hackathon`: joined a competition that requires a secret code (= hackathon / class event)
- **Community:** `n_discussions`, `n_comments`
- **Profile:** `FeatureX`, `FeatureY`, `country`, `country_missing`, `signup_day`, `signup_hour`, `days_observed`

```
[8]: counts = pd.crosstab(a0.User_ID, a0.family).reindex(columns=FAMILIES, fill_value=0)
counts.columns = ['n_' + c.lower().replace(' ', '_') for c in counts.columns]
activity = a0.groupby('User_ID').agg(n_events=('family', 'size'), n_active_days=('day',
↳ 'nunique'),
                                   n_distinct_actions=('family', 'nunique'),
↳ last_active_day=('day', 'max'),
                                   n_distinct_comps=('comp_id', 'nunique'))

first_seen = pd.concat([part[['Competition ID', 't']],
                        act.dropna(subset=['comp_id']).rename(columns={'comp_id':
↳ 'Competition ID'})[['Competition ID', 't']]]
                        ).groupby('Competition ID').t.min()
p0 = p0.assign(new_comp=(p0.t == p0['Competition ID'].map(first_seen)).astype(int),
               hackathon=p0['Competition ID'].map(comps.set_index('Comp_ID').SecretCode),
               sub_bucket=p0['Successful Submission Count'].str.extract(r'(\d+)',
↳ expand=False).astype(float))
competition = p0.groupby('User_ID').agg(n_comps_joined=('Competition ID', 'nunique'),
↳ max_sub_bucket=('sub_bucket', 'max'),
                                   n_new_comps_joined=('new_comp', 'sum'),
↳ joined_hackathon=('hackathon', 'max'))

df = (users.set_index('User_ID')
      .join(activity).join(counts).join(competition)
      .assign(n_discussions=d0.User_ID.value_counts(), n_comments=c0.User_ID.value_counts()))
zero_fill = [c for c in df.columns if c.startswith('n_')] + ['max_sub_bucket',
↳ 'joined_hackathon']
df[zero_fill] = df[zero_fill].fillna(0)

obs_end = act.groupby('month').day.max()           # last logged day per month (masked lengths;
↳ M4 log stops at day 22)
df['days_observed'] = (df.month.map(obs_end) - df.signup_day + 1).clip(lower=1)
```

```

df['days_since_last'] = (df.month.map(obs_end) - df.last_active_day).clip(lower=0).fillna(df.
    ↪ days_observed)
df['active_day_ratio'] = df.n_active_days / df.days_observed
df['signup_hour'] = df.time.str[:2].astype(int)
df['country_missing'] = df.Countries_ID.isna().astype(int)
top10 = df.Countries_ID.value_counts().index[:10]
df['country'] = np.select([df.Countries_ID.isin(top10), df.Countries_ID.isna()], [df.
    ↪ Countries_ID, 'Missing', 'Other'])
df = pd.get_dummies(df, columns=['country', 'FeatureY'], prefix=['cty', 'FY'])
df = df.drop(columns=['Countries_ID', 'time', 'Created At Year', 'month', 'last_active_day'])
print('modelling table:', df.shape, '| remaining NaNs:', int(df.isna().sum().sum()))

```

modelling table: (12413, 52) | remaining NaNs: 0

```

[9]: FEATURES = [c for c in df.columns if c not in ('Active', 'cohort')]
lab = df[df.cohort.isin(LABELLED)]
X, y = lab[FEATURES], lab['Active'] # <- inputs / output
X_test = df.loc[df.cohort == TEST_COHORT, FEATURES] # M4 cohort, label hidden
print(f'X: {X.shape}, y: {y.shape}, positive rate = {y.mean():.3f} | X_test: {X_test.shape}')
X[['n_events', 'n_active_days', 'n_comps_joined', 'n_created_submission', 'days_since_last',
    ↪ 'country_missing']].describe().T.round(2)

```

X: (9026, 50), y: (9026,), positive rate = 0.305 | X_test: (1382, 50)

```

[9]:

```

	count	mean	std	min	25%	50%	75%	max
n_events	9026.0	18.77	25.95	0.0	8.0	12.0	21.0	885.0
n_active_days	9026.0	1.60	1.70	0.0	1.0	1.0	2.0	21.0
n_comps_joined	9026.0	0.56	0.87	0.0	0.0	0.0	1.0	12.0
n_created_submission	9026.0	0.53	3.59	0.0	0.0	0.0	0.0	103.0
days_since_last	9026.0	11.72	9.10	0.0	4.0	10.0	18.0	31.0
country_missing	9026.0	0.39	0.49	0.0	0.0	0.0	1.0	1.0

3. Exploratory analysis — 3.1 retention differs strongly by cohort

```

[10]: r = lab.assign(cohort=lab.cohort.map(LABEL))
g = r.groupby('cohort').Active
rate = g.mean().reindex([LABEL[t] for t in LABELLED]); n_users = g.size().reindex(rate.index)
ci = 1.96 * np.sqrt(rate * (1 - rate) / n_users) # 95 % CI of a
    ↪ proportion

fig, axes = plt.subplots(1, 2, figsize=(13, 4.6), gridspec_kw={'width_ratios': [1, 1.3]})
ax = axes[0]
ax.bar(rate.index, rate.values, yerr=ci.values, capsize=4, width=0.62,
    color=[RED if c == 'M12' else BLUE for c in rate.index], error_kw={'lw': 1.2, 'ecolor':
    ↪ '#333333'})
for x, (v, e) in enumerate(zip(rate.values, ci.values)):
    ax.text(x, v + e + 0.015, f'{v:.0%}', ha='center', va='bottom', fontsize=12)
ax.yaxis.set_major_formatter(PercentFormatter(1)); ax.set_ylim(0, 0.95); ax.grid(axis='x',
    ↪ visible=False)
ax.set(title='Share active in month 2, by sign-up cohort', ylabel='P(active next month)',
    ↪ xlabel='Sign-up cohort')

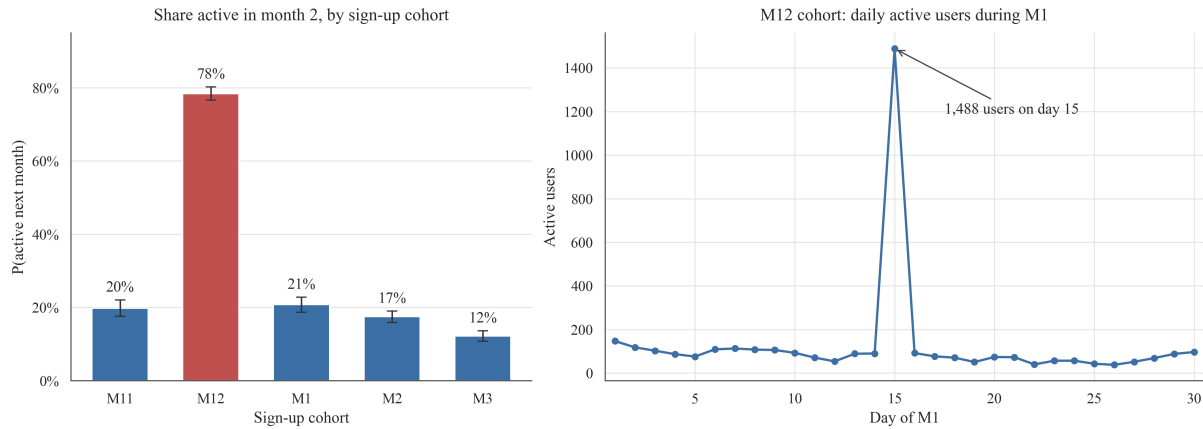
m12 = act.merge(users[['User_ID', 'cohort']], on='User_ID').query('cohort == 1 and t == 2')
dau = m12.groupby('day').User_ID.nunique()
ax = axes[1]

```

```

ax.plot(dau.index, dau.values, marker='o', ms=4.5, lw=1.8, color=BLUE)
pk = dau.idxmax()
ax.annotate(f'{dau[pk]:,} users on day {pk}', xy=(pk, dau[pk]), xytext=(pk + 2.5, dau[pk] * 0.
↪8),
          arrowprops={'arrowstyle': '->', 'color': '#444444', 'lw': 1}, fontsize=12)
ax.set(title='M12 cohort: daily active users during M1', ylabel='Active users', xlabel='Day_
↪of M1',
      xlim=(0.5, dau.index.max() + 0.5))
plt.show()

```



- 4 of 5 cohorts retain **12–21 %**, but the **M12 cohort retains 78 %**.
- M12 users' activity in M1 is one **spike on a single day (about 1 500 users on day 15)**, concentrated on competition ID_528W: a *secret-code hackathon* hosted in country X9GR, running from M12-28 to M2-3. M12 sign-ups were mostly **event participants**, whose “retention” was mechanically caused by the event.
- **Consequences for ML**
 1. Label prior is not stationary → a random train/test split would mix cohorts and look over-optimistic → **time-based validation**.
 2. The M12 cohort is not representative of the test cohort → we test **excluding it from training** (Section 4).
 3. Motivated two features: `joined_hackathon`, `n_new_comps_joined`.

3.2 First-month engagement vs. month-2 activity

```

[11]: normal = lab[lab.cohort != 1]                # typical cohorts

def rate_panel(ax, groups, y, color, xlabel, title, label_side='right'):
    # P(active next month) per bin with a 95 % CI; the bin size is shown under each tick
    g = y.groupby(groups, observed=True)
    p, n = g.mean(), g.size()
    ci = 1.96 * np.sqrt(p * (1 - p) / n)
    x = np.arange(len(p))
    ax.errorbar(x, p.values, yerr=ci.values, fmt='o-', color=color, ms=7, lw=1.6, capsize=4)
    for xi, v in zip(x, p.values):
        ax.annotate(f'{v:.0%}', (xi, v), textcoords='offset points', xytext=(10 if label_side_
↪== 'right' else -10, 0),
                  ha='left' if label_side == 'right' else 'right', va='center', fontsize=11)

```

```

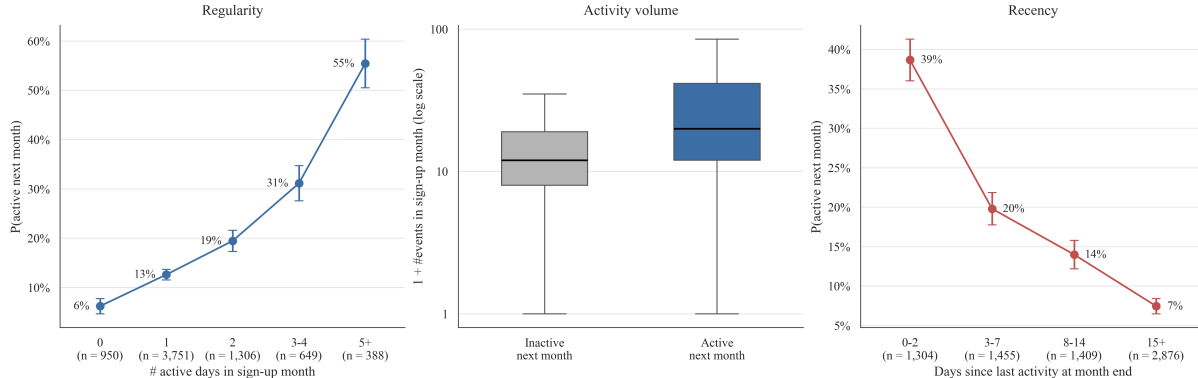
ax.set_xticks(x, [f'{k}\n(n = {m:,})' for k, m in zip(p.index.astype(str), n.values)])
ax.yaxis.set_major_formatter(PercentFormatter(1, decimals=0)); ax.grid(axis='x',
↪visible=False)
ax.set(xlabel=xlabel, ylabel='P(active next month)', title=title, xlim=(-0.6, len(p) - 0.
↪4))

fig, axes = plt.subplots(1, 3, figsize=(15.5, 4.9))
bins = pd.cut(normal.n_active_days, [-1, 0, 1, 2, 4, 31], labels=['0', '1', '2', '3-4', '5+'])
rate_panel(axes[0], bins, normal.Active, BLUE, '# active days in sign-up month',
↪'Regularity', label_side='left')

box = axes[1].boxplot([1 + normal.loc[normal.Active == k, 'n_events'] for k in (0, 1)],
↪widths=0.5, showfliers=False,
                        patch_artist=True, medianprops={'color': 'black', 'lw': 1.6},
                        whiskerprops={'color': '#555555'}, capprops={'color': '#555555'})
for patch, col in zip(box['boxes'], [GREY, BLUE]):
    patch.set(facecolor=col, edgecolor='#555555')
axes[1].set_yscale('log'); axes[1].yaxis.set_major_formatter(ScalarFormatter()); axes[1].
↪yaxis.set_minor_formatter(NullFormatter())
axes[1].set_xticks([1, 2], ['Inactive\nnext month', 'Active\nnext month']); axes[1].
↪grid(axis='x', visible=False)
axes[1].set(ylabel='1 + #events in sign-up month (log scale)', title='Activity volume')

rec = pd.cut(normal.days_since_last, [-1, 2, 7, 14, 31], labels=['0-2', '3-7', '8-14', '15+'])
rate_panel(axes[2], rec, normal.Active, RED, 'Days since last activity at month end',
↪'Recency')
plt.show()

```



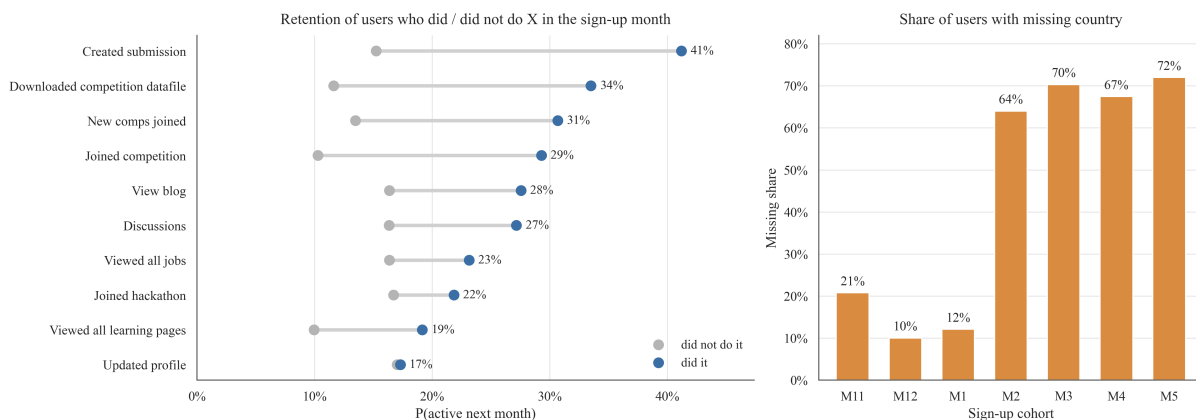
- Retention rises monotonically with the **number of active days**: 6 % (no logged activity) → 13 % (1 day) → 55 % (5+ days).
- **Recency** is at least as important as volume: users seen in the last 2 days of the month return 39 % of the time, vs 7 % for users silent for 15+ days.
- Distributions are extremely **right-skewed** (median user: a handful of events) → log-transform counts for the linear model; tree models are invariant.

3.3 Behaviour types and profile variables

```
[12]: beh = ['n_created_submission', 'n_downloaded_competition_datafile', 'n_joined_competition',
↳ 'n_viewed_all_learning_pages',
      'n_viewed_all_jobs', 'n_view_blog', 'n_updated_profile', 'n_discussions',
↳ 'joined_hackathon', 'n_new_comps_joined']
rates = pd.DataFrame({b[2:] if b.startswith('n_') else b: normal.groupby(normal[b] > 0).
↳ Active.mean() for b in beh}).T
rates.columns = ['did not', 'did']
rates = rates.sort_values('did')

fig, axes = plt.subplots(1, 2, figsize=(14.5, 5), gridspec_kw={'width_ratios': [1.45, 1]})
ax = axes[0]; yy = np.arange(len(rates))
ax.hlines(yy, rates['did not'], rates['did'], color='#D0D0D0', lw=3, zorder=1)
ax.scatter(rates['did not'], yy, color=GREY, s=70, zorder=2, label='did not do it')
ax.scatter(rates['did'], yy, color=BLUE, s=70, zorder=3, label='did it')
for yi, v in zip(yy, rates['did']):
    ax.annotate(f'{v:.0%}', (v, yi), textcoords='offset points', xytext=(8, 0), va='center',
↳ fontsize=11)
ax.set_yticks(yy, rates.index.str.replace('_', ' ').str.capitalize())
ax.xaxis.set_major_formatter(PercentFormatter(1, decimals=0)); ax.set_xlim(0, rates.values.
↳ max() * 1.15); ax.grid(axis='y', visible=False)
ax.set(xlabel='P(active next month)', title='Retention of users who did / did not do X in the
↳ sign-up month')
ax.legend(loc='lower right')

miss = users[users.cohort <= 6].groupby('cohort').apply(lambda g: g.Countries_ID.isna().
↳ mean()).rename(index=LABEL)
ax = axes[1]
ax.bar(miss.index, miss.values, color=ORANGE, width=0.62)
for x, v in enumerate(miss.values):
    ax.text(x, v + 0.012, f'{v:.0%}', ha='center', va='bottom', fontsize=11.5)
ax.yaxis.set_major_formatter(PercentFormatter(1)); ax.set_ylim(0, 0.82); ax.grid(axis='x',
↳ visible=False)
ax.set(title='Share of users with missing country', ylabel='Missing share', xlabel='Sign-up
↳ cohort')
plt.show()
```

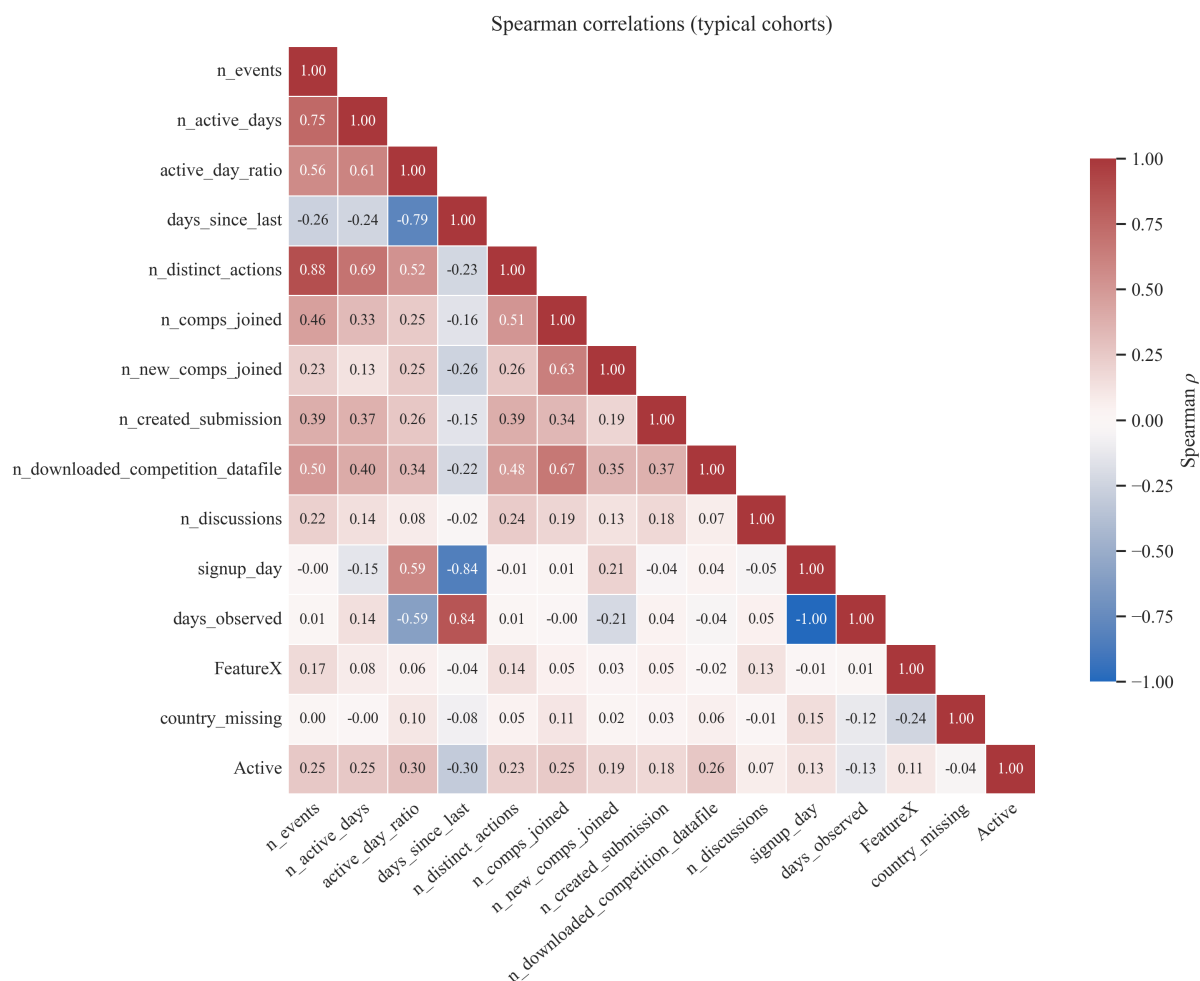


- *Doing* things (submitting, downloading data, joining competitions, posting) predicts retention much

better than *reading* (blog, jobs pages). → keep the behaviour-family breakdown instead of a single activity count.

- **Country is missing for 10–21 % of users up to M1 and 64–72 % from M2 on** — a change in the sign-up form, not user behaviour. `country_missing` is therefore a *time proxy* that can mislead a model trained on old cohorts → we check whether dropping country features helps under temporal validation.

```
[13]: corr_feats = ['n_events', 'n_active_days', 'active_day_ratio', 'days_since_last',
↳ 'n_distinct_actions', 'n_comps_joined',
↳ 'n_new_comps_joined', 'n_created_submission', 'n_downloaded_competition_datafile',
↳ 'n_discussions',
↳ 'signup_day', 'days_observed', 'FeatureX', 'country_missing', 'Active']
c = normal[corr_feats].corr(method='spearman')
fig, ax = plt.subplots(figsize=(11, 8.5))
sns.heatmap(c, mask=np.triu(np.ones_like(c, dtype=bool), 1), cmap='vlag', center=0, vmin=-1,
↳ vmax=1,
↳ annot=True, fmt='.2f', annot_kws={'size': 9.5}, linewidths=0.6,
↳ linecolor='white', square=True,
↳ cbar_kws={'shrink': 0.7, 'label': r'Spearman  $\rho$ '}, ax=ax)
ax.set_title('Spearman correlations (typical cohorts)'); ax.grid(False)
ax.tick_params(axis='both', length=0); ax.tick_params(axis='x', rotation=40)
plt.setp(ax.get_xticklabels(), ha='right', rotation_mode='anchor')
plt.show()
```



- Activity features are strongly **collinear** (ρ 0.6–0.9) $\rightarrow$ L2-regularised logistic regression; trees handle it natively.
- The strongest single correlates of **Active** are regularity (`n_active_days`) and recency (`days_since_last`, negative).
- `signup_day` and `days_observed` are mirror images ($\rho = -1$). Interestingly late sign-ups are *more* likely to be active next month ($\rho = +0.13$): their first sessions spill over into month 2. The model must see sign-up timing to interpret activity counts correctly.

4. Validation design and metric

Cross-validation — forward chaining by cohort (train on past cohorts, validate on the next one), because at deployment time the model is trained on past cohorts and applied to a new one; random K-fold would leak cohort-level information.

split	train cohorts	validate on
inner fold 1	M11 (+M12)	M1
inner fold 2	M11 (+M12), M1	M2
hold-out test	M11 (+M12), M1, M2	M3 (never used for tuning)

Hyper-parameters **and the final model family** are selected on the two inner folds only; the decision threshold is chosen on the inner out-of-fold predictions; the M3 cohort is scored **once** at the end.

Metric — F1 of the positive class (the competition metric): positives are only 12–21 %, so accuracy is useless (predict “all inactive” $\rightarrow$ 85 % accuracy, $F1 = 0$). F1 balances *precision* (don’t waste retention budget) and *recall* (don’t miss engaged users). For model selection we use **PR-AUC (average precision)**, which is threshold-free and focuses on the minority class.

dev: (7026, 50) hold-out (M3): (2000, 50) hold-out positive rate 0.121

Why not random K-fold? — a quick demonstration

```
[15]: demo = HistGradientBoostingClassifier(random_state=RNG)
p_rand = cross_val_predict(demo, X, y, cv=StratifiedKFold(5, shuffle=True, random_state=RNG),
    ↪method='predict_proba')[:, 1]
rows = [('random 5-fold (all cohorts mixed)', average_precision_score(y, p_rand))]
for tr, va in forward_folds(coh, val_cohorts=(2, 3, 4)):
    p = clone(demo).fit(X.iloc[tr], y.iloc[tr]).predict_proba(X.iloc[va])[:, 1]
    rows.append((f'forward: train < {LABEL[coh[va][0]]} → validate {LABEL[coh[va][0]]}',
    ↪average_precision_score(y.iloc[va], p)))
pd.DataFrame(rows, columns=['scheme', 'PR-AUC']).round(3)
```

```
[15]:
```

	scheme	PR-AUC
0	random 5-fold (all cohorts mixed)	0.849
1	forward: train < M1 → validate M1	0.296
2	forward: train < M2 → validate M2	0.469
3	forward: train < M3 → validate M3	0.464

Random K-fold reports PR-AUC ≈ 0.85 ; honest forward validation gives ≈ 0.3 –0.5. Part of the gap is the higher base rate when M12 is mixed in, but mostly the model **memorises cohort-specific signals**

(e.g. which hackathon, whether country is recorded) that do not transfer to a future cohort. Every number from here on uses forward validation.

5. Models and hyper-parameter search

model	why
Heuristic baseline — “active on $\geq k$ days”	what a product manager would do without ML
Logistic regression (log1p + scaling, L2, class-balanced)	interpretable, robust to shift, handles collinearity via regularisation
Random forest (class-balanced)	non-linear, robust, few assumptions
Histogram gradient boosting	state of the art for tabular data, native handling of skewed counts & interactions

Grid search (GridSearchCV with our forward folds, scoring = PR-AUC), then two data-level choices suggested by the EDA: (a) keep or drop the anomalous M12 cohort from training, (b) keep or drop country features.

```
[16]: log_lr = make_pipeline(FunctionTransformer(np.log1p), StandardScaler(),
                             LogisticRegression(max_iter=5000, class_weight='balanced'))
SEARCH = {
    'LogReg': (log_lr, {'logisticregression__C': [0.003, 0.01, 0.03, 0.1, 0.3, 1]}),
    'RandomForest': (RandomForestClassifier(n_estimators=400, class_weight='balanced_subsample',
    ↪ n_jobs=-1, random_state=RNG),
    ↪ {'max_depth': [6, 10, None], 'min_samples_leaf': [5, 20, 50]},
    ↪ 'max_features': ['sqrt', 0.4]),
    'GradBoost': (HistGradientBoostingClassifier(random_state=RNG),
    ↪ {'learning_rate': [0.03, 0.1], 'max_iter': [100, 300], 'max_leaf_nodes':
    ↪ [7, 15, 31],
    ↪ 'min_samples_leaf': [20, 80], 'l2_regularization': [0.0, 1.0]}),
}
NO_COUNTRY = [f for f in FEATURES if not f.startswith('cty_') and f != 'country_missing']

def search(name, drop_m12, feats):
    est, grid = SEARCH[name]
    folds = forward_folds(coh_dev, exclude=(1,) if drop_m12 else ())
    gs = GridSearchCV(est, grid, scoring='average_precision', cv=folds, n_jobs=-1,
    ↪ refit=False)
    gs.fit(X_dev[feats], y_dev)
    return gs.best_params_, gs.best_score_

results = []
for name in SEARCH:
    for drop_m12 in (False, True):
        for feat_set, feats in (('all', FEATURES), ('no country', NO_COUNTRY)):
            params, score = search(name, drop_m12, feats)
            results.append(dict(model=name, train_M12='dropped' if drop_m12 else 'kept',
    ↪ features=feat_set,
                                cv_PR_AUC=score, params=params))

res = pd.DataFrame(results)
res.pivot_table(index='model', columns=['train_M12', 'features'], values='cv_PR_AUC').round(3)
```

```
[16]: train_M12    dropped      kept
features      all no country  all no country
```

model				
GradBoost	0.499	0.508	0.425	0.425
LogReg	0.471	0.474	0.436	0.450
RandomForest	0.500	0.498	0.446	0.470

- **Dropping the M12 cohort from training helps every model** — the event-driven cohort teaches the wrong base rate and wrong patterns.
- Country features: small / inconsistent effect → we keep the simpler, more shift-robust variant when scores tie (see below).
- Selected configuration per model (best CV PR-AUC):

```
[17]: best = res.sort_values('cv_PR_AUC', ascending=False).groupby('model').head(1).
      ↪ set_index('model')
      for name, p in best.params.items():
          print(f'{name:>12}: ' + ', '.join(f"{k.split('__')[-1]}={v}" for k, v in p.items()))
      best[['train_M12', 'features', 'cv_PR_AUC']].round(3)
```

```
GradBoost: l2_regularization=1.0, learning_rate=0.03, max_iter=100,
max_leaf_nodes=7, min_samples_leaf=80
RandomForest: max_depth=6, max_features=0.4, min_samples_leaf=50
LogReg: C=0.003
```

```
[17]:
```

	train_M12	features	cv_PR_AUC
model			
GradBoost	dropped	no country	0.508
RandomForest	dropped	all	0.500
LogReg	dropped	no country	0.474

5.1 Threshold selection and final evaluation on the hold-out cohort (M3)

For each tuned model: out-of-fold probabilities on the inner folds → pick the threshold maximising F1 → refit on all development cohorts → score M3 once.

```
[18]: def f1_threshold(y_true, p):
      grid = np.arange(0.05, 0.96, 0.01)
      return grid[np.argmax([f1_score(y_true, p >= t) for t in grid])]

      def build(name, params):
          return clone(SEARCH[name][0]).set_params(**params)

      final, probs = [], {}
      for name, row in best.iterrows():
          feats = FEATURES if row.features == 'all' else NO_COUNTRY
          excl = (1,) if row.train_M12 == 'dropped' else ()
          oof_y, oof_p = [], []
          for tr, va in forward_folds(coh_dev, exclude=excl):
              m = build(name, row.params).fit(X_dev.iloc[tr][feats], y_dev.iloc[tr])
              oof_y.append(y_dev.iloc[va]); oof_p.append(m.predict_proba(X_dev.iloc[va][feats])[:, 1])
          ↪ 1])
          thr = f1_threshold(np.concatenate(oof_y), np.concatenate(oof_p))
          keep = ~np.isin(coh_dev, excl)
          model = build(name, row.params).fit(X_dev[keep][feats], y_dev[keep])
          p = model.predict_proba(X_hold[feats])[:, 1]; probs[name] = p
          final.append(dict(model=name, threshold=thr, F1=f1_score(y_hold, p >= thr),
          ↪ precision=precision_score(y_hold, p >= thr),
```

```

        recall=recall_score(y_hold, p >= thr), PR_AUC=average_precision_score(y_hold,
    p), ROC_AUC=roc_auc_score(y_hold, p)))
    best.loc[name, 'threshold'] = thr

# heuristic baseline: active on >= k days, k tuned on the same inner folds
oof = np.concatenate([va for _, va in forward_folds(coh_dev, exclude=(1,))])
k = max(range(1, 10), key=lambda k: f1_score(y_dev.iloc[oof], X_dev.n_active_days.iloc[oof]
    >= k))
pb = (X_hold.n_active_days >= k).astype(int)
final.append(dict(model=f'Baseline (active days >= {k})', threshold=np.nan,
    F1=f1_score(y_hold, pb), precision=precision_score(y_hold, pb),
        recall=recall_score(y_hold, pb), PR_AUC=np.nan, ROC_AUC=np.nan))
final.append(dict(model='Predict everyone active', threshold=np.nan, F1=f1_score(y_hold, np.
    ones_like(y_hold)),
        precision=y_hold.mean(), recall=1.0, PR_AUC=y_hold.mean(), ROC_AUC=0.5))
final = pd.DataFrame(final).set_index('model')
final.round(3)

```

```

[18]:
threshold      F1  precision  recall  PR_AUC
ROC_AUC
model
GradBoost      0.32  0.472      0.451  0.494  0.479
0.830
RandomForest   0.60  0.462      0.412  0.527  0.451
0.817
LogReg          0.61  0.484      0.433  0.547  0.482
0.820
Baseline (active days >= 2)   NaN  0.359      0.259  0.588      NaN
NaN
Predict everyone active      NaN  0.217      0.122  1.000  0.122
0.500

```

- **Final model = GradBoost** (highest CV PR-AUC 0.508, trained without M12, without country features). On the hold-out cohort it reaches **F1 = 0.47** (precision 0.45, recall 0.49, ROC-AUC 0.83).
- All three ML models are within ~0.02 F1 of each other on M3 (LogReg is marginally higher, 0.48) — differences of that size are within the noise of a 243-positive test cohort, so we keep the CV-selected model rather than cherry-picking on the hold-out.
- Every ML model beats the product rule-of-thumb (+0.11 F1) and the trivial “everyone active” rule (+0.25 F1).

```

[19]: METRICS = {'F1': '#3A6EA5', 'precision': '#9DBBDD', 'recall': '#E0A33B', 'PR_AUC': '#8C8C8C'}
MODEL_COLORS = {'GradBoost': '#C0504D', 'RandomForest': '#6A4C93', 'LogReg': '#2A9D8F'}

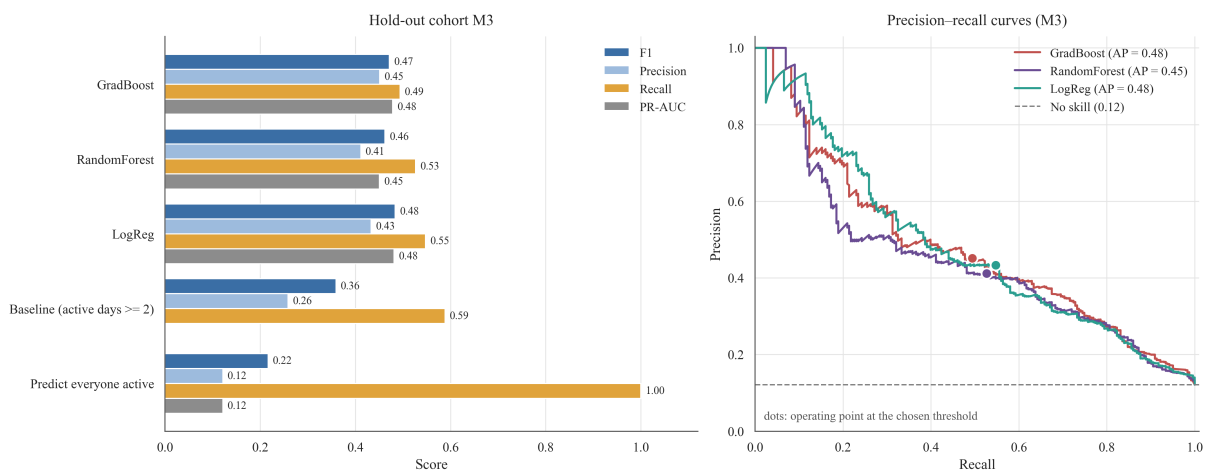
fig, axes = plt.subplots(1, 2, figsize=(14.5, 5.6), gridspec_kw={'width_ratios': [1.2, 1]})
ax = axes[0]; models = final.index.tolist(); yy = np.arange(len(models)); h = 0.2
for i, (m, col) in enumerate(METRICS.items()):
    vals = final[m].values.astype(float); ypos = yy + (i - 1.5) * h
    ax.barh(ypos, np.nan_to_num(vals), height=h, color=col, label=m.replace('_', '-').
    replace('precision', 'Precision').replace('recall', 'Recall'))
    for yp, v in zip(ypos, vals):
        if not np.isnan(v):
            ax.text(v + 0.01, yp, f'{v:.2f}', va='center', fontsize=9.5)
ax.set_yticks(yy, models); ax.invert_yaxis(); ax.set_xlim(0, 1.12); ax.grid(axis='y',
    visible=False)
ax.set(title='Hold-out cohort M3', xlabel='Score'); ax.legend(loc='upper right')

```

```

ax = axes[1]
for name, p in probs.items():
    col = MODEL_COLORS.get(name, BLUE)
    pr, rc, _ = precision_recall_curve(y_hold, p)
    ax.plot(rc, pr, color=col, lw=1.8, label=f'{name} (AP = {average_precision_score(y_hold,
    ↪p):.2f})')
    ax.plot(final.loc[name, 'recall'], final.loc[name, 'precision'], 'o', color=col, ms=9,
    ↪mec='white', mew=1.5)
ax.axhline(y_hold.mean(), ls='--', c='grey', lw=1.2, label=f'No skill ({y_hold.mean():.2f})')
ax.text(0.02, 0.03, 'dots: operating point at the chosen threshold', transform=ax.transAxes,
    ↪fontsize=10.5, color='#555555')
ax.set(xlabel='Recall', ylabel='Precision', title='Precision-recall curves (M3)', xlim=(0, 1.
    ↪01), ylim=(0, 1.03))
ax.legend(loc='upper right')
plt.show()

```



5.2 What drives the best model?

```

[20]: BEST = best.cv_PR_AUC.idxmax() # chosen on CV only - the hold-out is never used for
    ↪selection
row = best.loc[BEST]; feats = FEATURES if row.features == 'all' else NO_COUNTRY
excl = (1,) if row.train_M12 == 'dropped' else ()
best_model = build(BEST, row.params).fit(X_dev[~np.isin(coh_dev, excl)][feats], y_dev[~np.
    ↪isin(coh_dev, excl)])
pi = permutation_importance(best_model, X_hold[feats], y_hold, scoring='average_precision',
    ↪n_repeats=10, random_state=RNG, n_jobs=-1)
imp = pd.Series(pi.importances_mean, feats).sort_values(ascending=False).head(12)

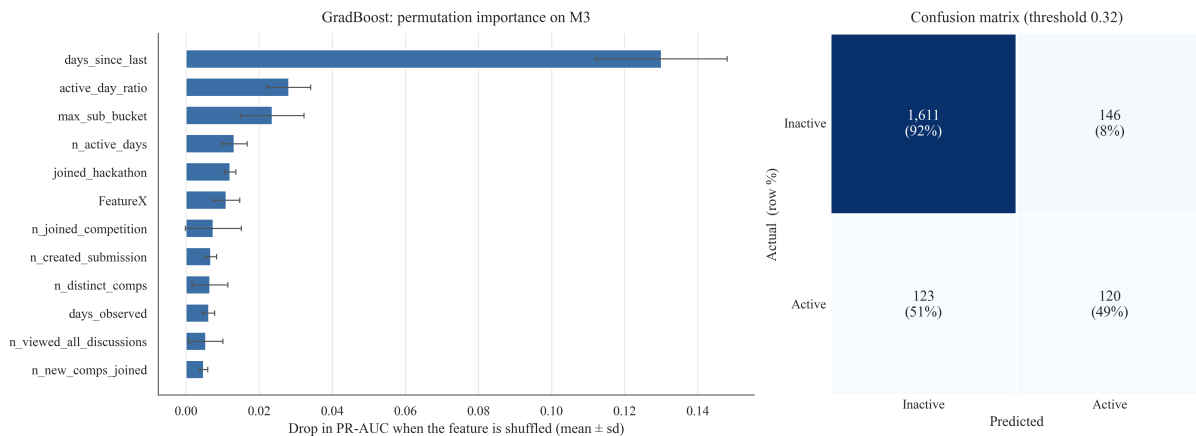
p_best = probs[BEST]; thr = best.loc[BEST, 'threshold']
imp_sd = pd.Series(pi.importances_std, feats)[imp.index]
fig, axes = plt.subplots(1, 2, figsize=(14.5, 5.2), gridspec_kw={'width_ratios': [1.6, 1]})
ax = axes[0]
ax.barh(imp.index[::-1], imp.values[::-1], xerr=imp_sd.values[::-1], color=BLUE, height=0.65,
    ↪error_kw={'lw': 1, 'ecolor': '#555555', 'capsize': 2.5})
ax.grid(axis='y', visible=False)

```

```

ax.set(title=f'{BEST}: permutation importance on M3', xlabel='Drop in PR-AUC when the feature_
↳is shuffled (mean ± sd)')
cm = confusion_matrix(y_hold, p_best >= thr)
ann = np.array([[f'{v:},}\n({v / cm[i].sum():.0%})' for v in cm[i]] for i in range(2)])
sns.heatmap(cm, annot=ann, fmt='', cmap='Blues', cbar=False, ax=axes[1], linewidths=1.5,
↳linecolor='white',
        annot_kws={'size': 13}, xticklabels=['Inactive', 'Active'],
↳yticklabels=['Inactive', 'Active'])
axes[1].set(title=f'Confusion matrix (threshold {thr:.2f})', xlabel='Predicted',
↳ylabel='Actual (row %)')
axes[1].tick_params(axis='both', length=0); axes[1].tick_params(axis='y', rotation=0);
↳axes[1].grid(False)
plt.show()

```



- **Recency** (`days_since_last`) is by far the most important signal, followed by **regularity** (`active_day_ratio`, `n_active_days`) and *doing* signals (successful-submission bucket, competitions joined, submissions, hackathon participation).
- Profile variables contribute little (only the masked `FeatureX` appears in the top 12; country was dropped by the search) — engagement is about **behaviour, not demographics**, which is good for fairness and actionability.
- At the F1-optimal threshold the model finds about half of the returning users with ~45 % precision ($\approx 3.7\times$ the 12 % base rate).

6. Real-world impact — who should Zindi target?

A retention campaign has a budget: contact the top- x % of users ranked by the model.

```

[21]: order = np.argsort(-p_best)
cum_pos = np.cumsum(y_hold.values[order]) / y_hold.sum()
share = np.arange(1, len(order) + 1) / len(order)
top20 = cum_pos[int(.2 * len(order)) - 1]
bottom50 = 1 - cum_pos[int(.5 * len(order)) - 1]
dec = pd.qcut(pd.Series(p_best).rank(method='first'), 10, labels=range(10, 0, -1))
lift = pd.Series(y_hold.values).groupby(dec).mean() / y_hold.mean()
lift.index = lift.index.astype(int); lift = lift.sort_index()

fig, axes = plt.subplots(1, 2, figsize=(14.5, 4.9))

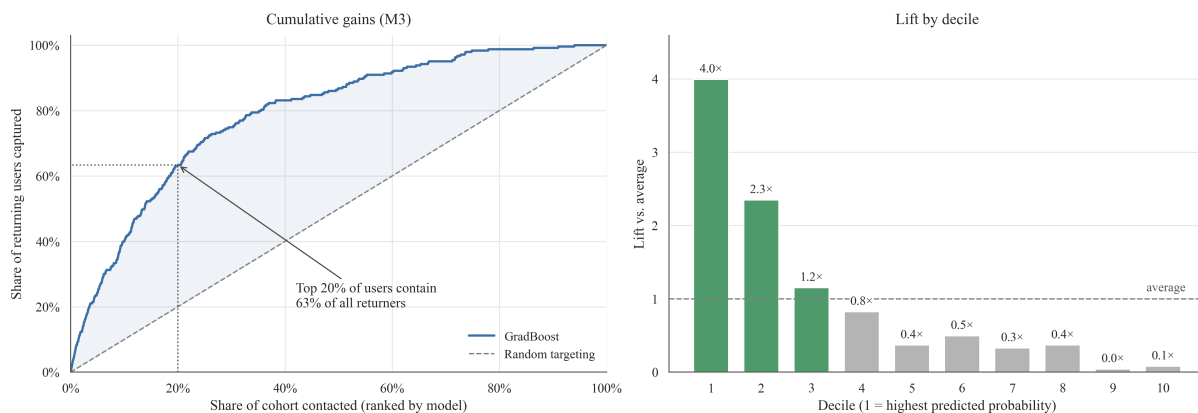
```

```

ax = axes[0]
ax.plot(share, cum_pos, color=BLUE, lw=2, label=BEST)
ax.plot([0, 1], [0, 1], '--', c='grey', lw=1.2, label='Random targeting')
ax.fill_between(share, share, cum_pos, color=BLUE, alpha=0.08)
ax.plot([0.2, 0.2, 0], [0, top20, top20], ':', color='#555555', lw=1.2)
ax.annotate(f'Top 20% of users contain\n{top20:.0%} of all returners', xy=(0.2, top20),
    ↪xytext=(0.42, 0.2),
    ↪arrowprops={'arrowstyle': '->', 'color': '#444444', 'lw': 1}, fontsize=12)
ax.xaxis.set_major_formatter(PercentFormatter(1)); ax.yaxis.set_major_formatter(PercentFormatter(1))
ax.set(xlabel='Share of cohort contacted (ranked by model)', ylabel='Share of returning users',
    ↪captured',
    ↪title='Cumulative gains (M3)', xlim=(0, 1), ylim=(0, 1.03))
ax.legend(loc='lower right')

ax = axes[1]
ax.bar(lift.index, lift.values, color=[GREEN if v >= 1 else GREY for v in lift.values],
    ↪width=0.7)
for x, v in zip(lift.index, lift.values):
    ax.text(x, v + 0.06, f'{v:.1f}x', ha='center', va='bottom', fontsize=11, zorder=3,
        ↪bbox={'facecolor': 'white', 'edgecolor': 'none', 'pad': 0.6})
ax.axhline(1, ls='--', c='grey', lw=1.2); ax.text(10.45, 1.07, 'average', ha='right',
    ↪va='bottom', fontsize=11, color='#555555')
ax.set_xticks(range(1, 11)); ax.set_ylim(0, lift.max() * 1.15); ax.grid(axis='x',
    ↪visible=False)
ax.set(xlabel='Decile (1 = highest predicted probability)', ylabel='Lift vs. average',
    ↪title='Lift by decile')
plt.show()
print(f'Top 20 % by score contain {top20:.0%} of all returning users; bottom 50 % contain
    ↪only {bottom50:.0%}.')

```



Top 20 % by score contain 63% of all returning users; bottom 50 % contain only 13%.

Reading the curves: the top decile is 4× more likely to return than average; the top 20 % of users by score contain 63 % of all returners, while the bottom half contains only 13 %.

Illustrative business case (assumptions stated explicitly — to be replaced by Zindi's real numbers):

- *Churn-prevention campaign:* the model's **bottom half** holds very few returners, so Zindi can concentrate a costly personal intervention (mentor call, tailored beginner competition) on the **middle**

deciles — users with some engagement but uncertain outcome — instead of the whole cohort.

- *Community / recruiting*: the **top decile** is several times more likely to be active than average → the natural shortlist for hackathon invitations and ambassador programmes.
- Example: 2 000 sign-ups/month, intervention cost 5 USD/user. Contacting everybody = 10 000 USD. Contacting the 40 % “uncertain” segment = 4 000 USD (−60 %) while still covering the users whose behaviour can plausibly be changed.
- The *causal* effect of an intervention cannot be estimated from these observational data → an **A/B test** (random hold-out within the targeted segment) is required before claiming revenue impact.

6.1 Predictions for the official test cohort (M4)

```
[22]: keep = ~np.isin(coh, excl)                                # final model: all labelled cohorts
      ↪ except excluded ones
final_model = build(BEST, row.params).fit(X[keep][feats], y[keep])
p_test = final_model.predict_proba(X_test[feats])[:, 1]
sub = pd.DataFrame({'User_ID_Next_month_Activity': X_test.index + '_Month_5', 'Active':
      ↪ (p_test >= thr).astype(int)})
sub = sample_sub[['User_ID_Next_month_Activity']].merge(sub, how='left').fillna({'Active':
      ↪ 0}).astype({'Active': int})
sub.to_csv('submission.csv', index=False)
print(f'{len(sub)} rows written to submission.csv; predicted active share = {sub.Active.
      ↪ mean():.3f}')
print(f'(M4 activity log ends on day {act[act.t == TEST_COHORT].day.max()} - late sign-ups
      ↪ have truncated features)')
sub.head()
```

1340 rows written to submission.csv; predicted active share = 0.109

(M4 activity log ends on day 22 - late sign-ups have truncated features)

```
[22]: User_ID_Next_month_Activity  Active
0      ID_4TOXNBGB_Month_5         0
1      ID_CHFTIP26_Month_5         0
2      ID_FU5GMWLQ_Month_5         0
3      ID_254TVBQP_Month_5         0
4      ID_4EN02VFE_Month_5         0
```

7. Conclusions

Findings

- The labelled training set had to be *reconstructed*: we de-anonymised the month order from the data itself and derived the target.
- **Recency and regularity of first-month activity** predict month-2 retention far better than demographics; *doing* (submitting, downloading data, joining) beats *browsing*.
- Retention is **non-stationary**: one hackathon made a cohort’s retention jump from ~15 % to 78 %; the sign-up form change made **country** missing for most later users. Both would silently break a naïvely validated model.

ML experiment critique

- Random K-fold would have overstated performance by ~2×; forward-chaining CV is the honest estimate.
- Excluding the event-driven cohort from training was the single most useful “data” decision; model choice mattered less than validation design.

- Limits: only 5 labelled cohorts (small, noisy fold estimates); masked features; M4 log truncated at day 22; retention of event cohorts is driven by an *external calendar* (hackathons) that the features only partially capture.

Did we solve the problem? Largely for *ranking*, partly for *exact classification*. On an unseen future cohort the model reaches $F1 = 0.47$ vs 0.36 for the best rule of thumb, and concentrates 63 % of returners in the top 20 % of users — enough to make targeted retention/community programmes several times cheaper than blanket campaigns. Absolute F1 stays moderate because many returns are triggered by future events (new competitions, hackathons) that are invisible in month 1.

Next steps: add the platform’s competition calendar (upcoming launches) as features; train on more cohorts; calibrate probabilities per cohort; validate business impact with an A/B-tested retention campaign.